

System Error: Where Big Tech Went Wrong and How We Can Reboot

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Today, the heroic figures are the disruptive and instantly wealthy innovators. Whereas once technologists brought with them countercultural visions of enhancing human capabilities, promoting liberty and equality, and spreading democracy, today the culture of Silicon Valley is about founder worship and the celebration of apolitical coders. This was a profound shift that technologists didn't notice or didn't want to acknowledge until they had to in the wake of the social and political fallout from technology's role in Brexit, the election of Trump, and the siege of the US Capitol.

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When you marry the engineers' mindset to the venture capitalists' profit motive, you get an obsession with scale.

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We are entering a new era in which the rules that govern technology will no longer be written by hackers or companies alone. They will reflect some push and pull among the companies that make things, the governments that oversee them, the consumers who use them, and the people who are affected by them.

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well-intentioned optimizers fail to measure all that is meaningful, and when their creative disruptions achieve great scale, they impose their values and decisions upon the rest of us.

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Our aim is to understand how the pursuit of optimization by technologists—often seen as a good in itself—can undermine the well-being of individuals and the health of democratic societies.

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As James Madison wrote in the Federalist Papers, "If Men were angels, no government would be necessary." When the stakes are high and the power of technology over our lives is enormous, relying on technologists alone, even ethical ones, is a mistake.

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Although democracies can strive to achieve the best outcomes for citizens, they can also serve as guardrails against the worst outcomes. Karl Popper, the twentieth-century Austrian philosopher, believed that the central problem of politics is not to decide who should rule: the people (a democracy), the wisest (a philosopher king or a beneficent technocrat), or the wealthiest (an oligarchy). The central problem of politics is how to organize political institutions so that terrible outcomes can be avoided and bad or incompetent rulers can be prevented from doing grave damage. This is the moment we find ourselves in as the impacts of technology overwhelm our society.

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the market is good at some things and not at others. It rewards profit without regard to social consequences. It prizes efficiency while ignoring other values. It celebrates domination. These priorities are encoded in the algorithms that power new technologies, the metrics that drive company strategy, and the regulatory environment that governs what companies may and may not do. But there are other ideals at stake: fairness, privacy, autonomy, equality, democracy, justice. As human beings, we treasure these concepts and protect them through the rules we've established to govern ourselves. Yet new technologies have put many of them at risk—sometimes in ways that are visible but often in ways that are not.

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We can't leave our technological future to engineers, venture capitalists, and politicians. This book lays out the dangers of leaving the optimizers in charge and empowers all of us to make the difficult decisions that will determine how technology transforms our society.

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In an age of advanced technology, inefficiency is the sin against the Holy Ghost. —Aldous Huxley, foreword to *Brave New World*, 1946

Stelle: 638

What we choose to optimize raises the issue of representational adequacy. In order to optimize some quantity (ticket cost, travel time) we need to have a way to represent that quantity mathematically. If we can't directly measure or represent that quantity, there's no way to create an optimization method to find out if it's doing better or worse.

Stelle: 652

A recent popular book, *Algorithms to Live By: The Computer Science of Human Decisions*, recommends the skills of a computer scientist as the basis for leading a better life: algorithmic insights as a form of wisdom.

Stelle: 676

The real worry is that giving priority to optimization can lead to focusing more on the methods than on the goals in question. If a particular problem to be solved is simply delivered to a software engineer without consideration or debate as to its value, we are stuck with the results of optimizing the goal.

Stelle: 683

The pioneering computer scientist Donald Knuth famously said that “premature optimization is the root of all evil.”

Stelle: 709

What responsibility does the technologist have for trying to ensure that a technology is used for good rather than bad? Such questions are becoming more and more urgent in tech companies as engineers increasingly grapple with the unexpected ways in which their work may be used.

Stelle: 715

It's worth emphasizing that whoever makes the choice of what to optimize is effectively deciding what problems are worth solving. The glaring lack of racial and gender diversity in the ranks of technologists and start-up founders means that these choices rest in the hands of a small group of people not representative of the wider world. No surprise that many new start-ups show a bias in favor of solving the problems of a privileged demographic.

Stelle: 742

Economists have long worried about the problem of proxies, especially in situations where employees receive incentives for meeting their targets. In such situations, employees will quickly orient themselves not toward the worthy end but toward the proxy. The metric becomes the goal, and the means justify the end. This is called Goodhart's Law, which states that when a measure becomes a target, it ceases to be a good measure. A common sequence would look like this: The boss says we must make progress toward a large and difficult-to-measure goal. The leaders of the company choose some proxies that seem to have a plausible connection to the goal. Employees are tasked with making progress on these proxies. Though they may recognize that the given proxy is a useful but imperfect measure for the goal, they begin to lose sight of the imperfections of the proxy and thereby of the good goal, too. Soon enough, the proxy becomes the only thing that matters.

Stelle: 772

"Success disasters" abound in Silicon Valley technologies as well. Facebook, YouTube, and Twitter have succeeded in connecting billions of people in a social network, but now that they have created a digital civic square, they and not the government have to grapple with the conflict between freedom of expression and the spread of misinformation and hate speech. The problems created by optimizing technologists arise not because their companies have failed but rather because they have succeeded profoundly at something and become powerful as a result.

Stelle: 827

The utopian notions of technology as a great equalizer have given way to dystopian stories of data breaches, surveillance capitalism, biased algorithms, and rampant misinformation.

Stelle: 856

In his book *Measure What Matters: How Google, Bono, and the Gates Foundation Rock the World with OKRs*, Doerr espoused the management principle of objectives and key results (OKRs), a concept originally developed by Andy Grove at Intel and now widely used at a number of technology companies, including Google, Twitter, and Uber. OKRs are the metrics that drive performance evaluations and corporate growth.

Stelle: 866

The marriage of technology and capital has come to define the "move fast and break things" culture of Silicon Valley. The countercultural notion of a free and uncontrolled cyberspace has given way to the new mantra of "blitzscaling," where companies grow as quickly as possible to grab a dominant market position, demonstrate hockey-stick growth to their investors, and lock in any potential network effects before competitors can respond.

Stelle: 941

The use of OKRs became the standard by which nearly everyone at Google was measured. As Doerr wrote, "the marriage of Google and OKRs was anything but random. It was a great impedance match, a seamless gene transcription into Google's messenger RNA. OKRs were an elastic, data-driven apparatus for a freewheeling, data-worshipping enterprise."

Stelle: 981

Doerr himself acknowledged that management systems such as OKRs can have their faults, writing, “Like any management system, OKRs may be executed well or badly.” He even affixed a warning label in his book, originally suggested by a Harvard Business School paper with the cheeky title “Goals Gone Wild”: “Goals may cause systematic problems in organizations due to narrow focus, unethical behavior, increased risk taking, decreased cooperation, and decreased motivation. Use care when applying goals in your organization.” What is missing is an explicit call for moral commitments to be an objective in themselves, rather than just a side consideration in the pursuit of some more directly measurable goal. The fact that it can be impossible to clearly measure the “social goodness” of an outcome means that it becomes difficult to factor it into the metric being optimized. It’s easier to just assume that watching more videos must be making users happier. Measuring watch time is straightforward; determining whether users are actually happier, more factually informed, or politically radicalized, is not.

Stelle: 1,016

When click-through rates increase on untruthful ads that question the integrity of election results without evidence or promote conspiracy theories about vaccines, revenues and market capitalizations may rise, but democracy and the well-being of hundreds of millions of people will suffer. Even when there is fallout for the companies responsible, it’s often short lived. Consider that Facebook’s stock price did not suffer any long-term damage from the Cambridge Analytica scandal in 2016. To the contrary, it has since soared. If the market rewards only revenue, where’s the incentive to protect democracy or other values we cherish?

Stelle: 1,074

Nicole Wong, who had served as vice president and deputy general counsel at Google before joining Twitter as legal director of products and eventually being appointed deputy chief technology officer of the United States in the Obama administration, discussed the emphasis on user engagement in online platforms such as YouTube. She argued for a “slow food” movement in technology in which, rather than valuing speed and engagement, platforms should seek to promote authenticity, accuracy, and context of content. In recounting her time at Google, she explained that the problems on platforms such as YouTube had become clear only once they scaled up, but perhaps slowing down the process of pushing products out the door might lead to more reflection about their potential impact and the ability to rethink the criteria that might actually indicate success.

Stelle: 1,091

John Doerr, speaking at the National Venture Capital Association in 2008, famously described how being a male nerd “correlates more with any other success factor that I’ve seen in the world’s greatest entrepreneurs. If you look at [Amazon founder Jeff] Bezos, or [Netscape founder Marc] Andreessen, [Yahoo cofounder] David Filo, the founders of Google, they all seem to be white, male, nerds who’ve dropped out of Harvard or Stanford and they absolutely have no social life.” But looking for patterns like this in who to fund also leads to inequities in how entrepreneurs might be evaluated by the VCs they are pitching to.

Stelle: 1,195

The market power of large players can translate into significant influence and leeway in the political sphere. As companies are threatened with the passage of new regulations, they learn to use their influence and money to resist regulation and shape policies. As Alvaro Bedoya, the executive director of the Center on Privacy and Technology at Georgetown University, told a reporter during the BIPA case, Facebook’s approach has been “If you sue us, it doesn’t apply to us; if you say it does apply to us, we’ll try to change the law.”

Stelle: 1,263

A recent Stanford study systematically mapped the libertarian attitudes of tech leaders—a distinctive mix of views that are socially progressive and economically conservative—who turned out to be even more hostile to regulation than were run-of-the-mill millionaires. The researchers concluded that this embrace of libertarian views manifests itself early, even among undergraduate computer science majors.

Stelle: 1,268

Bill Foster, a physicist and one of the few members of Congress with a PhD, noted in 2012 that only about 4 percent of members in Congress had technical backgrounds. “Most members of Congress don’t know enough about science and technology to know what questions to ask,” said former congressman Rush Holt of New Jersey, “and so they don’t know what answers they’re missing.”

Stelle: 1,285

Are we prepared to trade off other things, such as unfettered technological development, to ensure democratic oversight of such innovations?

Stelle: 1,375

It’s clear that our traditional choreography of regulation has its limits. In the words of the Nobel Prize-winning economist Paul M. Romer, “Finding good rules is not a one-time event.” Our rules need to evolve quickly as new technologies arrive. They need to be responsive to increases in scale. And they need to be robust to the opportunistic actions of individuals and companies who will try to undermine them. Romer called this Myron’s Law after another Nobel Prize-winning economist, Myron Scholes, who once commented in a seminar, “Asymptotically, any finite tax code collects zero revenue.” By that he meant that clever people will find ways to work around anything that is fixed. The bottom line is that our rules need to be as dynamic as the technologies they are designed to regulate.

Stelle: 1,420

The decision to allow internet innovators to operate outside the public utility regulatory framework that governed telephone companies was the accelerant that sped up our arrival at the present moment.

Stelle: 1,423

This approach also left unanswered critical questions that have come back to haunt us today, especially as we focus less on access and more on what happens to the content that moves across our communications networks. For example, must internet service providers and platforms treat all content equally as they decide what traffic moves and at what speed, just as telephone companies are forced to do under the old common carrier rules? Or can they make decisions based on what they like or don’t like, their preferences reflecting editorial discretion, profit concerns, or some other motivation? The stakes here are enormous. These issues, bundled under the moniker of “net neutrality,” are ultimately questions about how we are to manage market concentration and market power. The battle over net neutrality is ultimately a fight about whether your internet service provider has the right to speed up or slow down content passing through its network in order to make more money or favor particular providers.

Stelle: 1,457

with concern growing about tech’s negative consequences and the extraordinary degree of market power in these industries, the era in which network and platform companies have made their own rules is coming to an end. The fiction that innovation simply outruns democracy is no longer tenable. Democracy shares the blame, and we need to figure out how to do better.

Stelle: 1,510

The economist Bryan Caplan summed up the core problem in a memorable yet disturbing turn of phrase: “In my view, democracy fails because it does what voters want.”

Stelle: 1,547

In the words of the international affairs expert Tom Nichols, a passionate advocate for expertise in our public life, “Experts can only offer alternatives. They cannot, however, make choices about values . . . voters must engage those issues and decide what they value most, and therefore what they want done.”

Stelle: 1,626

If the decisions we confront in regulating technology do not have obvious answers—and most do not, for reasonable people can disagree how much free speech is good and what decisions should be kept out of the hands of robots—the idea that we will build some sort of shared and definitive view about

how all good things and competing values can be reconciled is fanciful, especially in a world in which new technologies are coming online so quickly.

Stelle: 1,638

If Winston Churchill was right that democracy is nothing more than “the worst form of Government except for all those other forms,” it is worth focusing on the minimal yet fundamental task at which democracies seem to excel: avoiding catastrophic outcomes and striving for stability and resilience to unpredictable shocks. Because even if we can’t agree on exactly the kind of society we want to live in, we can typically come to a consensus on the worst outcomes we want to avoid, such as imposing harm on individuals, being cruel to disadvantaged populations, and creating second-class citizens.

Stelle: 1,648

in a world in which nothing is more fundamental than access to food, politicians know that a failure to avoid famine is sufficient cause for defeat at the ballot box. So they work harder—and harder than those in nondemocratic regimes—to avoid this worst-case scenario. Democracy, it turns out, is a kind of man-made technology that eliminates famine.

Stelle: 1,660

The right approach, in Popper’s view, is to prepare from the beginning for the possibility of bad government and to ask: “How can we so organize our political institutions that bad or incompetent rulers can be prevented from doing too much damage?” This shifts our attention away from trying to find the best, most expert leaders to creating rules and institutions that enable us to remove bad leaders and reward good leaders when we have them. The bottom line is that we need good rules, not simply good rulers. And good rules are not a fixed point once they are deduced; they need to adapt to changing social and economic conditions, including technological innovations. And so democracies, which welcome a clash of competing interests and permit the revisiting and revising of questions of policy, will respond by updating rules when it is obvious that current conditions produce harm; when people suffer needlessly or unfairly; when a fire in a sweatshop awakens us to a long-simmering problem.

Stelle: 1,673

We need an alternative model of what we want from our politics—not utopian social engineering, as Plato imagined, but “piecemeal engineering,” in Popper’s words. Or in the words of another twentieth-century philosopher, Judith Shklar, we want a democratic society that rejects “a summum bonum toward which all political agents should strive,” but that does begin with “a summum malum, which all of us know and would avoid if only we could.” Without a blueprint for where we want to end up—because such a thing is impossible to achieve—we must focus instead on identifying and mitigating the harms and suffering we want to avoid. At this task, democracies have generally excelled: avoiding mass starvation; preventing nuclear war; eliminating extreme poverty and suffering.

Stelle: 1,769

as job applicants become aware that automated tools are used for screening their résumés, there’s no end to the ways they can game the system. Say you’re an applicant and you know your résumé will be analyzed by a machine. You could submit an electronic copy with some extra text in a white font on a white background at the bottom of the page. Since the text is white, it’s invisible to any human who reads it online or prints a copy. But the automated résumé-screening tool will scan and process all those terms just as if they appeared in black ink on a white page. That “extra” text might just include every phrase that you think might give you an edge for the position you’re applying for, including the entire list of required and desired applicant attributes in the original job posting. To stand out even more, why not add the names of an assortment of prestigious universities: Harvard, Oxford, Berkeley. Or signal something about your social position by including extracurricular activities such as “equestrian club” and “squash team.” If you think that sounds far-fetched, think again. These are all examples shared with us by current and former students. This kind of gaming has been going on for years among the tech savvy, and it’s only likely to get worse as algorithmic decision-making tools become more broadly used.

Stelle: 1,786

Russian president Vladimir Putin declared, “Whoever becomes the leader in [artificial intelligence] will become the ruler of the world.”

Stelle: 1,877

Most people think there is a common understanding of fair and unfair. Yet fairness resists easy definition.

Stelle: 1,895

Fairness is an idea that we apply to individuals as well as to groups. When we think about a hiring algorithm, we can construe fairness as a property of individuals, so that all applicants with identical skills and experiences get the same predictive score when they are reviewed by the algorithm. We can also construe fairness as a property of groups so that the percentage of minority-group members who are rated as hireable is the same as that of the majority group. Both conceptions of fairness seem important and reasonable. Yet we can’t easily adopt both at the same time when designing algorithms.

Stelle: 1,902

Fairness may not be consistently reducible to math, and it may vary somewhat across different social contexts, but this does not mean that it is subjective. We can still find our way and make use of this philosophical ideal.

Stelle: 1,923

When thinking about algorithmic decision-making, it is helpful to distinguish between two kinds of fairness: substantive and procedural. Substantive fairness focuses on the outcomes of some decision. Procedural fairness focuses on the process that generates the outcome. If the process is deemed fair, we don’t have to concern ourselves with the outcome. Algorithms that are fair will involve both substantive and procedural considerations.

Stelle: 1,927

For fairness, the most important issue is determining what counts as a morally relevant consideration in the decision-making process.

Stelle: 1,960

The advance of algorithmic decision-making is rapidly outpacing our ability to agree on what we want these systems to achieve.

Stelle: 1,962

Governments are getting into the game, including in areas you might have thought should be off limits to machines, such as how to allocate critical social services, whether to remove a child from a home, and what to teach children in school. In each case, someone has to make a decision about what the

algorithm should optimize and how fairness is to be achieved—and these decisions are virtually invisible.

Stelle: 2,034

Cathy O’Neil, a former mathematics professor turned data scientist, emphasizes this in her seminal book *Weapons of Math Destruction*, writing that algorithmic decision-making models “tended to punish the poor and the oppressed in our society, while making the rich richer.”

Stelle: 2,075

The only realistic strategy for engineers addressing bias is to systematically measure and take steps to mitigate bias and to be explicit about what conception of fairness they are optimizing for.

Stelle: 2,150

Reports point to three key ingredients in the governance of algorithmic decision-making. The first is transparency. [..]

The second key ingredient is auditability. Whenever possible, algorithms should be independently tested and validated, with the results made publicly available, including with explicit checks on bias. [..] Expertise is needed, and organizing and deploying this expertise will be one of the costs associated with harnessing the efficiency gains of these systems. [..]

The final ingredient is a commitment to due process. Individuals and groups should know how they may challenge decisions informed by an automated tool as well as the procedure for doing so and the response time after a request or complaint is lodged.

Stelle: 2,174

Though there is broad agreement that transparency, auditability, and due process are the principles needed to govern algorithmic systems, it shouldn’t surprise anyone that the devil is in the details.

Stelle: 2,187

Many of the “deep learning” models that are built today include many millions of parameters. Understanding how every one of them was affecting the model’s decision-making would be too complicated to figure out. But the engineers that built the model based its architecture on their understanding of the problem they were trying to solve. We can require the engineers to produce a high-level description of the model that would lay out their understanding of what it was supposed to encode. This could be augmented by various types of sensitivity analyses, tested on different gender, racial, and socioeconomic groups to see if they are treated similarly along dimensions that should not impact the decision-making process.

Stelle: 2,322

Despite the public concern of government surveillance programs and calls for greater regulation, information collection by private companies received little scrutiny by comparison. Indeed, until quite recently, the private sector has had scant regulation with respect to the information it can collect and process from the users of its products or systems. Companies such as Google and Facebook have been collecting troves of user interaction information—search queries, web pages visited through their platforms, friend connections, and “likes,” among others—for the purpose of honing their ad-targeting capabilities. The ability to optimize ad targeting with surgical precision quickly became the engine of the enormous revenues at these companies. Google alone generated more than \$130 billion in advertising revenue in 2019 as a result of employing large-scale machine learning algorithms to take its ad-targeting capabilities to unparalleled levels.

Stelle: 2,452

Thinking about a digital panopticon makes painfully clear what is lost in a world when everything is observed: we lose our privacy, and with that, we undermine our freedom, diminish the possibility of intimacy, and compromise our capacity to control what others know about us. As Bentham so clearly understood, surveillance is a means of social control, an opportunity for others to exercise power over us.

Stelle: 2,700

The “privacy paradox”: There is an enormous gap between what individuals say about the importance of privacy and their actual behavior. It’s not just a problem of aligning what you want with how you act. In light of so many intangible or invisible consequences, it is genuinely difficult to determine how much privacy we actually want.

Stelle: 2,796

GDPR is substantially different from the US Notice and Choice approach. Instead of relying on individual users to make informed judgments about the terms and conditions they are willing to accept, the GDPR adds distinct legal criteria under which personal data can be collected and creates a set of affirmative consumer rights to which every company serving EU residents is bound.

Stelle: 2,988

A future is coming in which we will have less agency over our lives, the benefits and burdens of automation will be unevenly distributed, and we may even find ourselves replaced by the technologies we can now create.

Stelle: 3,042

Until machines are capable of defining their own goals, the choices of the problems we want to solve with these technologies—what goals are worthy to pursue—are still ours.

Stelle: 3,158

According to one recent count, more than eighty distinct AI ethics documents have been released in the past decade, with the overwhelming majority appearing since 2016.

Stelle: 3,165

An ethics framework that states as general principles that an AI system must promote justice and be fair, bias free, privacy protecting, and accountable does not really provide much guidance. As we've seen in earlier chapters, the interesting and relevant issues arise when we ask deeper and harder questions about what fairness means, whether it is possible for an algorithm to encode it, and if it is, what price in predictive power we should be willing to pay to ensure it. Of course privacy is valuable, but an AI principle that announces a firm commitment to it tells us nothing about how we should balance the trade-offs between privacy and security.

Stelle: 3,172

There are two key questions: First, under what circumstances do massive advances in AI threaten to undermine human agency and perhaps challenge the very idea of what it is to be human? Second, what does increasing automation do to the material well-being of workers whose jobs are displaced or transformed?

Stelle: 3,198

Nozick was posing a simple and fundamental question, familiar not only to philosophers but to everyone who has wondered about our purpose in life: Is happiness, specifically the experience of being happy, the only important thing in life? In answering that question in the negative, he was taking aim at utilitarianism, the philosophical creed first developed by Jeremy Bentham. Utilitarianism holds that the ultimate good in life—the summum bonum—is happiness, understood as the experience of pleasure, and that the morally correct action for any individual is that which maximizes happiness for all, the greatest good of the greatest number. The implicit moral task of technologists, then, is to identify the best means to maximize happiness. In saying that anyone reading about the experience machine would decline to plug in, Nozick was pointing to the significance of knowing that our own agency, our own effort and talent, are connected to the experiences we have in life. “We want to be importantly connected to reality, not to live in a delusion,” he said. What's troubling about the experience machine is that apart from deciding to plug in, our own efforts bear no causal connection to our experience. True happiness can be achieved only when we bring about our own pleasure or well-being, not when you get its simulacrum for free. The most incredible virtual reality device might have some appeal, but would you really want to plug in forever, even if you were guaranteed the maximal flow of pleasure you could imagine? For many people, the answer is no. We generate meaning in life not merely from the felt experience of pleasure, pain, or anything in between but also from knowing that our actions, our very intentions and efforts, imperfect though they may be, are directly and causally connected to what we experience.

Stelle: 3,228

Smart machines might increasingly outperform humans and deliver greater productivity, but at bottom, we can't automate human flourishing.

Stelle: 3,280

Attaching a value to human agency isn't straightforward. This is a challenge Amartya Sen took up in his effort to reimagine the purpose of economic development. He took as his starting point Aristotle's famous dictum that “wealth is clearly not the good we are seeking, since it is merely useful, for getting something else.” Sen didn't want to measure only whether countries were rich or poor. He wanted to focus attention on what material wealth brings—whether people in any country have the freedom to live as they would like, to develop and deploy their interests and talents. To Sen, economic development is first and foremost about attaining freedom and unlocking the exercise of human capabilities. This perspective challenges the common conception of well-being. It forces us to consider not only wealth but the key ingredients that enable people to lead a full and meaningful life, such as education; a long life expectancy; access to clean water, food, and health care; political freedom; and civil liberties.

Stelle: 3,395

The tax code in the US exacerbates the situation by taxing labor at rates of roughly 25 percent but equipment and software at less than 5 percent, an effective subsidy to corporations that purchase machines and software that automate the workplace.

Stelle: 3,399

Iyad Rahwan, a Syrian Australian computer scientist and director of the Max Planck Institute for

Human Development in Berlin, has been working for years at the intersection of computer science and society. He believes that we need to move beyond the idea of “humans-in-the-loop” as we confront the potential uses of AI that have widespread societal implications. In this context, when further automation implicates all of us, we need to ensure that society’s values are reflected in the ways that machines are programmed and ultimately deployed. He calls this keeping “society-in-the-loop.”

Stelle: 3,440

If we think of AI as an alternative to human labor and cognition, the costs to our agency and well-being will be dramatic. But if political leaders use research-and-development investments and tax incentives to drive AI in ways that create new high-productivity tasks for human labor, it could become a new engine of productivity and growth.

Stelle: 3,475

Jason Furman, the chairman of President Obama’s Council of Economic Advisers, noted in a speech in 2016, “The issue is not that automation will render the vast majority of the population unemployable. Instead, it is that workers will either lack the skills or the ability to successfully match with the good, high-paying jobs created by automation.”

Stelle: 3,575

We now face a trilemma, a tension among three important values. The first is the value of free expression, the individual right to speak and be heard without censorship, which brings with it the benefits of a broad and diverse marketplace of ideas. Yet in the digital age a strong commitment to free speech puts a second value—democracy itself—at risk. The decision by tech companies to permit nearly unconstrained user-generated content in the name of freedom of expression has yielded interference in elections in many democratic societies by hostile foreigners and standing misinformation and disinformation campaigns on issues of all stripes. Should the political posts in the 2016 US election by Macedonian teenagers and Russia’s Internet Research Agency be protected by free speech norms? Should it matter if the identical material is circulated by US citizens rather than foreign agents? Should the speech of elected leaders or candidates for public office be treated differently than the speech of ordinary citizens? Just as important, we must confront the algorithmic sorting of users into filter bubbles that contribute to growing polarization, extremism, and decreasing social trust, all of which threaten the health of democracy. An unwavering commitment to free speech in the digital age can thwart a third value: individual dignity.

Stelle: 3,609

The online world has fundamentally transformed the very nature of free expression.

Stelle: 3,625

In the wake of the 2016 US election, BuzzFeed News reported that the top fake election news stories “generated more total engagement on Facebook than top election stories from 19 major news outlets combined.”

Stelle: 3,749

The success of any democratic society requires a healthy information ecosystem of educated and informed voters, opportunities for reasoned exchange and deliberation, the ability to separate fact from fiction, and trust between fellow citizens and between citizens and their leaders.

Stelle: 3,789

Democratic societies strive to protect individual dignity, as evidenced by human rights documents that often begin with a declaration of the dignity of all persons. Such protections account for the limits placed on free expression in the form of hate speech laws in many different countries. Notably, the United States is an outlier with a virtual absence of hate speech regulation.

Stelle: 3,813

As one recent analysis concluded, “Even if most political exchanges on social media take place among people with similar ideas, cross-cutting interactions are more frequent than commonly believed, exposure to diverse news is higher than through other types of media, and ranking algorithms do not have a large impact on the ideological balance of news consumption.” This is because social media seems to expand people’s interactions with individuals outside their closest social circles—coworkers, relatives living in other places, acquaintances—who often share different, and more ideologically diverse, information.

Stelle: 3,818

This evidence sits uncomfortably with the popular perception that social media produces nothing but echo chambers that polarize the population. To reconcile these facts, it is important to recognize that a lot of what happens on social media involves a small minority of highly active and visibly partisan individuals. For example, a small number of very frequent users were responsible for the overwhelming majority of hyperpartisan content on Twitter ahead of the 2012 election. Another study demonstrated that a majority of people rely primarily on centrist websites for their news, and the minority who visit the most partisan websites are active news consumers and visit many other kinds of sites as well.

Stelle: 3,830

Matthew Gentzkow, tackled this question in two revealing ways. First, for every age cohort in the United States since 1996, polarization was measured in eight different ways, ranging from the extent of straight-ticket voting to how people reported feeling about members of the other political party. The striking finding was that polarization has increased most among the groups that are least likely to use the internet and social media—people in the oldest age cohorts. Based on this evidence, it’s hard to conclude that social media itself is the major driver of polarization.

Stelle: 3,927

According to the February 2021 report, Facebook found 99 percent of content violating its “Adult Nudity and Sexual Activity” standards before users report it. For “Suicide and Self-Injury,” that number dropped to 92 percent. And for “Bullying and Harassment,” it was only about 50 percent. This variability in performance reflects the subjectivity and nuance of the material being assessed.

Stelle: 3,950

Even assuming that most objectionable content can be identified quickly, the most formidable challenge—a contest between competing values—arrives when we ask how that content should be handled. Nathaniel Persily, our colleague at Stanford, has offered a taxonomy of “Ds” that provides a wide array of options: deletion, demotion, disclosure, delay, dilution, diversion, deterrence, and digital literacy.

Stelle: 4,040

It does not make sense for the future of a healthy public sphere to rest entirely in the hands of a small number of powerful companies. Governments must get involved—and in so doing balance the importance of free expression against other essential goals.

Stelle: 4,100

According to a 2021 Pew survey, four of every ten Americans and a majority of young adults have experienced online harassment. The experience falls disproportionately on minority and marginalized groups, with a large majority of gays, lesbians, and bisexuals reporting that they have been targeted with online abuse.

Stelle: 4,162

There are definitely reasons we should consider tech platforms publishers of content. Unlike the telephone company, which simply hosts a connection between two speakers, the companies are engaged in active curation of what users see and hear through algorithms tuned to maximize engagement.

Stelle: 4,167

Two additional things are different now, both of which matter for the prospects of building an effective regulatory regime for internet platforms. The first is the level of partisan polarization. It’s difficult to imagine parties on opposite ends of the political spectrum coming to agreement on what constitutes fake news and misinformation, when hateful language becomes incitement to violence, and whether a curation process is politically neutral. The second is that legislators historically justified the regulation of broadcasting in terms of the physical limits of airspace that created spectrum scarcity. Because there wasn’t enough broadcast spectrum to enable a fully competitive market, broadcasters were expected to serve the public interest as well as their own commercial interests. That led to television practices that many of us took for granted, such as balanced coverage of political issues and content relevant to local communities. Though internet companies dominate the new channels of communication, their market dominance isn’t a function of scarcity, and it could in principle be challenged by other companies.

Stelle: 4,278

Only if we acknowledge technology’s power to shape our hearts and minds, and our collective beliefs and behaviors, will the discourses of governance shift from fatalistic determinism to the emancipation of self-determination. —Sheila Jasanoff, *The Ethics of Invention*, 2016

Stelle: 4,313

To give a sense of scale, the training data for GPT-3 is nearly 45 terabytes in size, or more than four times the estimated size of all the printed material in the Library of Congress in 2000.

Stelle: 4,374

To fight for your rights in high-stakes decisions, you need to understand whether an algorithm is involved. In contexts such as being denied a mortgage, losing access to social services, or encountering the criminal justice system, you may have a right to seek more transparency into the decision-making process, and that can include determining if and how an algorithm was used. Indeed, a growing number of lawyers are finding success at uncovering the use of algorithmic decision-making and successfully challenging unjust results in the courts.

Stelle: 4,391

We can't—and shouldn't have to—rely on personal action alone to confront the disruptions caused by big tech. As we've argued throughout this book, we need to bring our collective action to bear if we want technology to respect the wider and richer set of values we want to uphold.

Stelle: 4,403

When problems are systemic, the solutions cannot rely only on individual action.

Stelle: 4,404

The combination of the technologists' optimization mindset, the aspiration to maximize profit and scale, and the market dominance of a few companies defines the core problem. We have allowed the visions of technologists and their revolutionary innovations to disrupt not just marketplaces but the values that we hold dear and are core to the healthy functioning of democratic societies. What we confront today in big tech's disruptions is not a question about pop-up boxes concerning our privacy choices or whether to delete Facebook. Systemic problems require system-wide solutions. And such solutions are the traditional province of government, not consumer response; collective, not individual action.

Stelle: 4,417

It's time that citizens engage in a vigorous debate about the values we want technology to promote, as opposed to settling for the values that technology and the small group of people who produce it impose on us.

Stelle: 4,433

The AI researcher Joy Buolamwini founded the Algorithmic Justice League to call attention to the often-malign effects of algorithms on marginalized populations. Tech workers are organizing within their own companies to bring social protections to gig workers. In the wake of the livestreamed Christchurch massacre, nongovernmental organizations around the world banded together with governments to create the Christchurch Principles, an effort to promote social media governance that better supports democracy and human rights. Universities in the United States have launched a public interest technology consortium, creating new pathways for young people to work on addressing social problems with technology.

Stelle: 4,461

“Americans share every movement and sentiment with Facebook and Google, yet we seem reluctant to allow the Department of Health and Human Services to monitor patient encounters, as Taiwan does, to track disease and determine what medical tests and treatments to order.”

Stelle: 4,464

Building government expertise in technology, especially in light of the growing calls by politicians and citizens alike to rein in the power of big tech, would be a step in the right direction.

Stelle: 4,467

Creating an alternative future that can engage all of us in the tensions and trade-offs surfaced by new technologies requires progress on three fronts: cultivating a greater appreciation and understanding of ethical issues among technologists, checking corporate power, and empowering citizens and democratic institutions to govern technology instead of passively allowing technology and technologists to govern us.

Stelle: 4,508

Today, medical practice and research are structured by a thick institutional web of professional norms, legal codes, state licensure bodies, federal agencies, and human rights doctrines.

Stelle: 4,515

The result is that individuals who interact with the medical establishment, whether through clinical care or research, can trust that doctors are held to common standards of training and the interests of the patient are foremost. They can trust that there are serious efforts to test and regulate the availability of pharmaceuticals, so that drugs available over the counter or via prescription have been vetted by professionals. In the span of a century, the experience of being a doctor or a patient was transformed through the advancement and institutionalization of medical ethics.

Stelle: 4,521

The reforms in medicine took decades, often in response to scandals that aroused public outrage.

Stelle: 4,528

If a bridge or building collapses because of mistakes made by professional engineers, companies are legally liable for damages and individuals can have their licenses to practice revoked. Software engineers bear almost no such responsibility. A clause, for example, buried in the license terms for Microsoft Excel provides a broad shield to Microsoft and its engineers from legal liability if it should turn out that the spreadsheet software contains coding errors that lead to miscalculations.

Stelle: 4,535

The ACM task force ultimately rejected the idea of licensing requirements. “The licensing of software engineers as PEs [professional engineers],” the committee concluded, “at best would be ignored and at worst would be damaging to our field. It would have no or negligible effect on safety.” A sister organization, the Institute of Electrical and Electronics Engineers (IEEE), had a different view and sought to create a professional licensing exam. After several years of development, the exam was launched in a voluntary phase in 2013. But that effort also foundered, due mainly, it seems, to lack of uptake from graduating students, as the licensing is not required for industrial practice. By 2018, the exam had been administered five times to a total of eighty-one candidates. The IEEE shuttered the project.

Stelle: 4,549

A robust infusion of ethics into the culture of software engineering requires efforts on three major fronts. First, we need to broaden the practice of what is called “value-based design.” [..]

Second, professional bodies such as the ACM and IEEE should inject a dose of steroids into a renewed conversation about professional norms, an enhanced code of ethics, and potential licensure. [..]

Finally, we need an overhaul of the way we teach young software engineers and aspiring tech entrepreneurs. Just as young biology and medical school students take courses in bioethics and draw upon a deep well of research in this area, so, too, should computer science departments develop ambitious and interdisciplinary new courses that draw upon a young but growing base of scholarship on the ethical and social dimensions of technology.

Stelle: 4,598

The idea is simple: to create a new generation of civic-minded technologists and policy makers. Modeled on the emergence of public interest law—an innovation that transformed legal education with the goal of preparing young lawyers to work in nonprofit organizations and the public sector—the ambition is big.

Stelle: 4,607

There’s no substitute for the inclusion of diverse perspectives if we want to grapple seriously with the competing values at stake in the design of new technologies.

Stelle: 4,641

The major US tech companies can no longer ignore this shifting landscape. Publicly, many of them are embracing the role of government as regulator, welcoming public deliberation about new laws and policies—all the while knowing that partisan polarization and gridlock will make change extremely unlikely. But just in case, behind the scenes, they are fighting these lawsuits and lobbying furiously to ensure that any new laws and regulations do not harm their bottom lines or dominant market positions.

Stelle: 4,660

The Nobel Prize–winning economist Joseph Stiglitz compared the business elite to a “dieter who would rather do anything to lose weight than actually eat less.” From Stiglitz’s perspective, business leaders would prefer to do anything other than “fundamentally question the rules of the game—or even alter their own behavior to reduce the harm of the existing distorted, inefficient, and unfair rules.”

Stelle: 4,672

An agenda to limit the power of big tech companies has three key components. The first is addressing the huge power imbalance between companies and consumers when it comes to control over users’ personal data.

Stelle: 4,688

The second component is giving greater voice in companies to those who are likely to be hurt by technological change.

Stelle: 4,698

The third component is an assertive effort to constrain the market dominance of the major tech companies. This means cracking down on monopolistic behavior and restricting anti-competitive mergers and acquisitions.

Stelle: 4,715

The polarization and legislative gridlock in so many democratic societies make it difficult—if not impossible—to have reasoned discussions about how best to balance competing values. But there was a time not long ago when the US government had a world-renowned scientific advisory body, which was copied by other countries.

Stelle: 4,718

Jack Gibbons, a folksy physicist from Tennessee, was the director for more than a decade of a little-known congressional agency, the Office of Technology Assessment. The OTA was born in 1972 at a time of growing public concern over pollution, nuclear energy, pesticides, and other hazards related to technological change. That was almost a decade after the publication of *Silent Spring*, the book that jump-started the environmental movement and focused public attention on the ubiquitous, fast-growing new technologies that promised great benefits and also posed great risks.

Stelle: 4,727

One distinctive feature of OTA reports, beyond their incisive technical analysis, was their commitment to providing a range of policy options without advocating for a specific one. That allowed policy makers to benefit from technical input and advice but make the difficult political choices themselves.

Stelle: 4,747

The Netherlands Office of Technology Assessment (NOTA) made a significant improvement on the US model by seeking to move beyond relying just on scientific experts. NOTA also incorporates citizen deliberation. Audrey Tang's efforts in vTaiwan aim for a similar form of citizenship empowerment.

Stelle: 4,764

The creation of “digital services” in both the United Kingdom and the United States that have been successful at recruiting strong technical talent from Silicon Valley and around the world proves that many technologists will enthusiastically serve in the public interest if given the opportunity to work on significant projects that can benefit millions of people. What this requires is a large-scale effort to recruit what former White House official Christopher Kirchhoff calls “tech teammates” through flexible mechanisms that can pull in top-flight talent with cutting-edge expertise and industry experience outside of the formal civil service channels.

Stelle: 4,780

We need a fresh approach to regulation. Innovation in the tech sector is both disruptive and unpredictable; regulatory structures are rigid and reactive. Regulatory overhauls often lag technological change by years, playing catch-up as the accumulating evidence of harmful or unintended consequences becomes too difficult to ignore. The slow pace of regulation has been good for technology companies, giving them almost free rein to experiment, test, and scale up without attention to the consequences. But what society needs is a more responsive approach to regulation that will enable us to try out new policy frameworks and learn about their effects before locking in a long-term strategy. Experts call this “adaptive regulation,” and although it seems promising in principle, it can be very difficult to put into practice.

Stelle: 4,791

The idea is to create living laboratories where regulators can observe the real-world effects of particular regulatory models. Fintech has been the major beneficiary of this approach, with firms able to test new offerings on consumers in real markets while regulators observe and evaluate their potential benefits and harms.

Stelle: 4,815

As we balance the promise and peril of new technologies in our own societies, we will need to think about how our policy choices and regulatory approaches interact with—and in fact, can be actively coordinated with—other countries that share similar values. Though each country will make its own choices, we can't lose sight of the importance of forging common rules for the digital realm. Otherwise, we could end up in a situation in which China's preference for state control will displace our long-standing commitment to an open internet, robust competition, and set of meaningful protections for digital rights.

Stelle: 4,820

One common assumption about technology and geopolitics is that the world stage is a contest among China's digital authoritarianism, America's digital innovation, and Europe's focus on regulation. This dynamic is what drives many tech leaders to warn about the "Chinese alternative" if regulation were to stifle innovation. But the choice between innovation and regulation is a false dichotomy. Unless democracies assert greater collective voice over technology policy, the choice we face could be between global technology companies that do not place the interests of individuals or democracies first and the authoritarian model of technology governance offered by China.

Stelle: 4,832

Democracy is itself a kind of technology, a design for social problem solving whose chief virtues are its defense of individual rights, empowerment of citizens' voices, and adaptability to ever-changing social conditions. Though it is fragile, with its current vulnerabilities on stark display, its roots go back centuries. It has proven resilient to any number of challenges in the past. The regulation of our technological future will be its next defining challenge.